

VS Code Remote Development using Tailscale (Office-PC ↔ Home-PC)

Objective

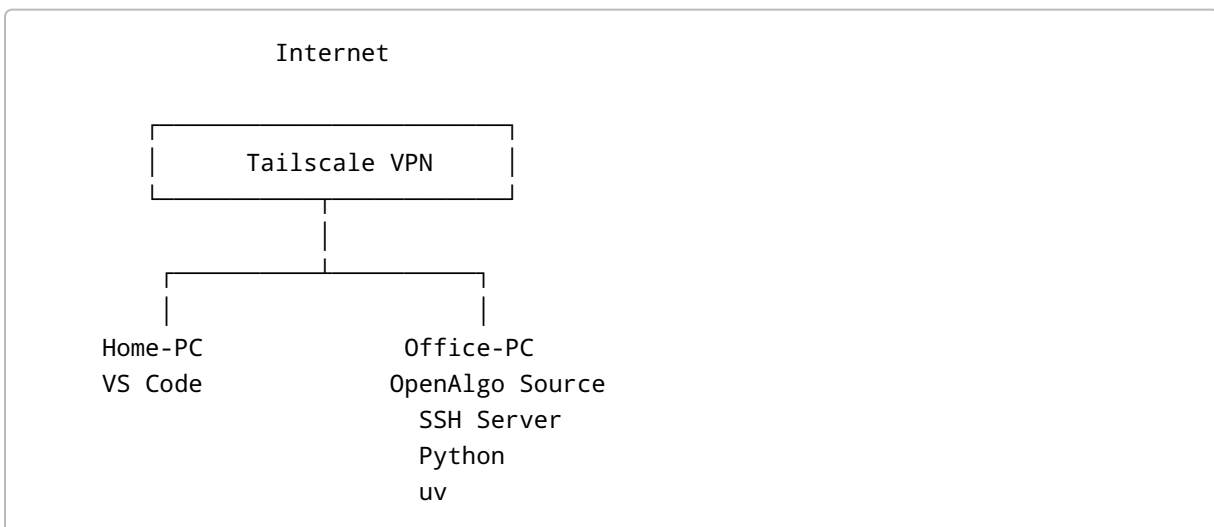
Develop and edit your **OpenAlgo** project running on the **Office-PC** from the **Home-PC** using **VS Code Remote SSH**.

This approach allows you to:

- Edit source code directly on the Office-PC
- Run terminals on the Office-PC
- Execute `uv run app.py` on the Office-PC
- Debug Python remotely
- Use Git on the Office-PC
- Avoid copying files between computers

All communication occurs securely over **Tailscale**.

Final Architecture



VS Code runs on the Home-PC.

All code execution happens on the Office-PC.

Prerequisites

Office-PC

- Windows
- OpenAlgo project
- Python
- uv
- Tailscale installed
- Logged into Tailscale

Home-PC

- Windows
 - VS Code
 - Tailscale installed
 - Logged into the same Tailscale account
-

Information to Note

Office-PC

Computer Name

DESKTOP-MB033IR

Windows Username

admin

This was obtained using

```
echo %env:USERNAME%
```

Output

admin

Important

`admin` is **your actual Windows account username**, obtained from:

```
echo %env:USERNAME
```

It is **not** a generic placeholder or example username.

On another Windows installation this value could be different (for example `john`, `administrator`, or `shreyas`). Always use the value returned by **your own** Office-PC.

Tailscale IP

```
100.108.179.50
```

Obtained using

```
tailscale status
```

Step 1 — Verify Tailscale

Office-PC

Run

```
tailscale status
```

Expected

```
100.108.179.50    desktop-mbo33ir
```

Home-PC

Run

```
tailscale status
```

You should see the Office-PC listed.

Step 2 — Install OpenSSH Server (Office-PC)

Open **PowerShell as Administrator**

Check whether OpenSSH Server is installed

```
Get-Service sshd
```

If Installed

Expected

Status	Name	DisplayName
Running	sshd	OpenSSH SSH Server

Proceed to Step 4.

If NOT Installed

Run

```
Add-WindowsCapability -Online -Name OpenSSH.Server~~~~0.0.1.0
```

Start the service

```
Start-Service sshd
```

Enable automatic startup

```
Set-Service -Name sshd -StartupType Automatic
```

Step 3 — Configure Windows Firewall

Open PowerShell as Administrator

Run

```
New-NetFirewallRule `
  -Name sshd `
  -DisplayName "OpenSSH Server" `
  -Enabled True `
  -Direction Inbound `
  -Protocol TCP `
  -Action Allow `
  -LocalPort 22
```

Verify

```
Get-NetFirewallRule sshd
```

Step 4 — Verify SSH Server

Run

```
netstat -ano | findstr :22
```

Expected

TCP	0.0.0.0:22	LISTENING
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If you do not see this,

SSH is not running.

Step 5 — Test SSH Connection

From the Home-PC

Open PowerShell

Run

```
ssh admin@100.108.179.50
```

where

```
admin
```

is the value returned by

```
echo %env:USERNAME
```

on the Office-PC.

First Connection

You will see

```
The authenticity of host cannot be established.  
Are you sure you want to continue connecting?
```

Type

```
yes
```

Enter the Windows password for the Office-PC user.

Windows SSH uses the **Windows account password**, not the Windows Hello PIN.

Expected prompt

```
admin@DESKTOP-MB033IR
```

SSH is now working.

Step 6 — Install VS Code Extension

On Home-PC

Open VS Code

Install extension

```
Remote - SSH
```

Published by Microsoft.

Step 7 — Configure SSH Host

Open Command Palette

```
Ctrl + Shift + P
```

Select

```
Remote-SSH: Open SSH Configuration File
```

Open

```
~/.ssh/config
```

Add

```
Host office-pc
```

```
HostName 100.108.179.50
```

```
User admin
```

where

```
admin
```

is the Office-PC Windows username returned by

```
echo $env:USERNAME
```

Save the file.

Step 8 — Connect from VS Code

Press

```
Ctrl + Shift + P
```

Select

```
Remote-SSH: Connect to Host
```

Choose

```
office-pc
```

VS Code will

- connect over SSH
- install the VS Code Server on the Office-PC (first time only)
- reconnect automatically

Step 9 — Open OpenAlgo

Inside VS Code

File

Open Folder

Browse to

C:\Users\admin\Downloads\ai tools\openalgo_platform\openalgo

(or wherever your OpenAlgo project resides)

Now

- Explorer shows Office-PC files.
 - Terminal runs on Office-PC.
 - Python executes on Office-PC.
-

Verify Everything

Verify SSH

From Home-PC

```
ssh admin@100.108.179.50
```

Expected

```
admin@DESKTOP-MB033IR
```

Verify VS Code

Open terminal

Run

```
hostname
```

Expected

```
DESKTOP-MB033IR
```

Run

```
whoami
```

Expected

```
desktop-mbo33ir\admin
```

Run

```
pwd
```

Should show the OpenAlgo project directory on the Office-PC.

Run

```
python --version
```

Should display the Python installed on the Office-PC.

Run

```
uv --version
```

Should display the Office-PC's `uv` version.

Starting OpenAlgo

Inside the VS Code terminal

```
uv run app.py
```

OpenAlgo now runs entirely on the Office-PC.

Accessing the Dashboard

Since the Office-PC is already configured with Tailscale,

you can open

```
http://100.108.179.50:5000
```

from

- Home-PC
- Mobile

while developing inside VS Code.

Workflow

```
Start Office-PC
```

```
↓
```

```
Tailscale starts
```

```
↓
```

```
OpenSSH starts
```

```
↓
```

```
OpenAlgo project available
```

↓

Home-PC

↓

Open VS Code

↓

Remote SSH

↓

Connect to office-pc

↓

Open OpenAlgo folder

↓

Edit Code

↓

Run

uv run app.py

↓

Test

<http://100.108.179.50:5000>

↓

Continue development

Troubleshooting

SSH Connection Refused

Check

```
Get-Service sshd
```

Service should be

```
Running
```

SSH Times Out

Verify

```
tailscale status
```

Both Office-PC and Home-PC should be online.

Verify firewall

```
netstat -ano | findstr :22
```

Should show

```
0.0.0.0:22 LISTENING
```

VS Code Cannot Connect

Verify SSH manually

```
ssh admin@100.108.179.50
```

If SSH works,

VS Code will also work.

Authentication Failed

Remember

SSH uses the

Windows account password

NOT

Windows Hello PIN.

Useful Commands

Office-PC

Current Username

```
echo %env:USERNAME
```

Current Computer Name

```
echo %env:COMPUTERNAME
```

Current Logged-in User

```
whoami
```

Verify SSH Service

```
Get-Service sshd
```

Verify SSH Port

```
netstat -ano | findstr :22
```

Verify Tailscale

```
tailscale status
```

Home-PC

SSH Test

```
ssh admin@100.108.179.50
```

Open VS Code

```
code .
```

Connect

```
Remote-SSH → office-pc
```

Advantages of This Setup

- No file synchronization required.
- Source code remains only on the Office-PC.
- Full IntelliSense and debugging.
- Terminal executes on the Office-PC.
- Git operations occur on the Office-PC.
- Secure encrypted communication via Tailscale.
- No port forwarding.
- No public IP.
- No domain name required.
- Accessible from anywhere with Tailscale connectivity.